Network Layer Case Study Report: The BGP Route Leak

Course: Computer Networks (CSA0712)

Topic: Network Layer – BGP Route Leak and Routing Failure

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2.Executive Summary:

A mid-sized company depends on an Internet Service Provider (ISP) for external Internet connectivity. Suddenly, users experience very slow Internet access, and some users are unable to access the company's corporate website.

The main clue is obtained from a traceroute, which shows packets repeatedly moving between two different Autonomous Systems (ASes). This indicates that the packets are following an incorrect routing path instead of reaching their intended destination.

The core problem is a BGP route leak. BGP (Border Gateway Protocol) is the routing protocol responsible for exchanging reachability information between Autonomous Systems on the Internet. In this case, an incorrect or improperly propagated route advertisement causes other networks to select an invalid path. As a result, traffic may be redirected through an inappropriate AS and can enter a routing loop.

The recommended solution is to identify the incorrect BGP advertisement by examining the BGP routing tables, withdraw or filter the invalid route, and verify that the correct route is being selected. Route filtering, prefix validation, and proper BGP policies should also be implemented to prevent similar incidents.

3. Problem Statement & Topology

3.1 Context

The company uses an ISP to connect its internal network to the Internet. Under normal conditions, Internet traffic should travel through the company's ISP toward the appropriate destination networks.

However, the company suddenly experiences:

- Very slow Internet connectivity.
- Some websites becoming unreachable.
- Increased packet delay.
- Repeated hops in traceroute.
- Traffic appearing to move between two Autonomous Systems repeatedly.

The traceroute result provides an important clue because repeated AS paths can indicate a routing loop caused by an incorrect BGP route advertisement.

3.2 Problem Statement

The network is experiencing an Internet connectivity failure caused by an invalid BGP route advertisement. The incorrect advertisement causes routers in different Autonomous Systems to select an inappropriate path, resulting in traffic being redirected between ASes.

The main objective is to identify the invalid BGP route, remove or filter the incorrect advertisement, restore the correct routing path, and verify normal connectivity.

3.3 Network Topology

The simplified topology is:

Corporate Network → Company Edge Router → ISP Router → AS-B → Destination Network

During the failure, the traffic may follow:

Company → ISP AS-A → AS-B → AS-A → AS-B → Destination

This repeated movement between AS-A and AS-B indicates a possible routing loop.

Topology components:

- Corporate LAN
- Company Edge Router
- ISP Router
- Autonomous System AS-A
- Autonomous System AS-B
- Destination Network
- BGP routing relationship between ASes

3.4 Scope

This case study focuses on the **Network Layer**, particularly:

- BGP routing
- Autonomous Systems
- IP prefix advertisements
- BGP route selection
- Routing loops
- Route filtering
- Traceroute-based troubleshooting

4. Technical Analysis & Troubleshooting

4.1 Evidence

The first troubleshooting step is to use **ping and traceroute** to determine where connectivity is failing.

For example:

C:\> tracert corporate-website.com

A problematic traceroute may show repeated hops:

1 Company Router

2 ISP Router

3 AS-B Router

4 AS-A Router

5 AS-B Router

6 AS-A Router

7 AS-B Router

...

The repeated appearance of the same networks indicates that packets are not progressing toward the destination.

This behavior suggests a routing loop or an incorrect route advertisement.

4.2 Root Cause

The failed protocol is **BGP (Border Gateway Protocol)**.

BGP exchanges IP prefix reachability information between Autonomous Systems. A route leak occurs when a network advertises a route to another network in a way that violates the intended routing policy.

For example:

AS-A advertises Prefix X to AS-B

AS-B propagates Prefix X incorrectly

AS-A receives an invalid route for Prefix X

Routers may then select the incorrect route according to their BGP decision process.

Consequently, packets may be forwarded through an inappropriate path and repeatedly circulate between Autonomous Systems.

4.3 How to Inspect the Routing Tables

On a Cisco router, the following commands can be used:

show ip route

This displays the router's IP routing table.

To specifically inspect BGP routes:

show ip bgp

To examine a particular destination prefix:

show ip bgp <destination-prefix>

For example:

show ip bgp 203.0.113.0

The output can be examined to determine:

- Which AS advertised the prefix.
- Which next hop was selected.
- The AS-PATH associated with the route.
- Whether multiple paths exist.
- Whether an unexpected AS appears in the path.

The command:

show ip bgp neighbors

can also be used to inspect BGP neighbor relationships.

For more detailed information about a BGP route:

show ip bgp <destination-prefix>

The **AS-PATH** is particularly important because it shows the sequence of Autonomous Systems through which the route has been advertised.

4.4 Protocols Involved

BGP

BGP is the main protocol involved in this failure. It exchanges routing information between Autonomous Systems and determines the best available path.

An incorrect BGP advertisement can cause:

- Incorrect routing.
- Traffic redirection.
- Routing loops.
- Increased latency.
- Packet loss.
- Website unreachability.

ICMP

ICMP is involved when testing connectivity using:

ping

and when routers generate responses used by:

traceroute

ICMP does not cause the BGP route leak; it helps identify the resulting connectivity problem.

Traceroute

Traceroute is a troubleshooting tool rather than a routing protocol. It helps identify the sequence of Layer 3 hops taken by packets.

Repeated hops between ASes provide evidence of a possible routing loop.

5. Proposed Solution & Implementation Plan

5.1 Resolution

The solution is to identify and remove the invalid BGP route advertisement.

The following steps can be followed.

Step 1: Test Connectivity

Use:

ping <destination-IP>

Then run:

tracert <destination>

or, on a Cisco device:

traceroute <destination-IP>

Step 2: Check the Routing Table

Run:

show ip route

This determines which route the router is currently using.

Step 3: Examine BGP Routes

Run:

show ip bgp

Look for the affected destination prefix.

Then use:

show ip bgp <destination-prefix>

to inspect the available BGP paths and AS-PATH.

Step 4: Identify the Invalid Advertisement

Compare the expected AS path with the path displayed by BGP.

An unexpected AS or an inappropriate route advertisement can identify the source of the route leak.

Step 5: Withdraw or Filter the Invalid Route

The incorrect advertisement should be withdrawn or blocked using appropriate BGP policy controls.

For example, prefix filtering can be configured using an access list and route policy.

Example:

ip prefix-list BLOCK-INVALID seq 5 deny 203.0.113.0/24

The exact configuration depends on the network design and the router vendor.

Step 6: Clear the Incorrect BGP Route if Required

After correcting the BGP policy, the affected route may need to be refreshed.

clear ip bgp <neighbor-IP> soft

A soft reset is preferred when possible because it avoids unnecessarily terminating the BGP session.

Step 7: Verify the Correct Route

Run:

show ip bgp <destination-prefix>

and:

show ip route <destination-IP>

The correct next hop and AS-PATH should now be selected.

5.2 Risk Assessment

Changing BGP policies can temporarily affect Internet connectivity. Incorrect filtering could accidentally block legitimate routes.

Therefore:

- Changes should be tested before deployment.
- Configuration backups should be maintained.
- Route filters should be carefully reviewed.
- Soft BGP resets should be preferred where appropriate.
- A rollback configuration should be prepared.
- Changes should be monitored after implementation.

6. Verification and Conclusion

6.1 Testing

After implementing the solution, connectivity should be tested again.

Ping Test

ping <destination-IP>

Successful replies indicate that IP connectivity has been restored.

Traceroute Test

traceroute <destination-IP>

The repeated AS-A → AS-B → AS-A pattern should no longer appear.

Routing Table Verification

show ip route

The router should contain the correct route to the destination.

BGP Verification

show ip bgp

The invalid route should no longer be selected.

The AS-PATH should show the expected routing path.

6.2 Conclusion

The case study demonstrates how a BGP route leak can cause serious Internet connectivity problems. The incorrect advertisement causes routers in different Autonomous Systems to select an inappropriate path, potentially resulting in routing loops and unreachable destinations.

- The primary protocol responsible for the failure is **BGP**. The problem can be identified using traceroute and BGP routing-table analysis. Cisco commands such as show ip route, show ip bgp, and show ip bgp <prefix> help network administrators identify the invalid path advertisement.
- The incident also demonstrates the importance of BGP route filtering, proper routing policies, prefix validation, and continuous monitoring. These measures improve network-layer resilience and reduce the possibility of future routing failures.

7. References & Appendices

References

1. Cisco Networking Academy, *Cisco Networking Essentials*.
2. Cisco IOS Documentation – BGP Configuration and Verification.
3. RFC 4271 – *A Border Gateway Protocol 4 (BGP-4)*.
4. RFC 7908 – *Problem Definition and Classification of BGP Route Leaks*.
5. Kurose, J. F. and Ross, K. W., *Computer Networking: A Top-Down Approach*.

Appendix A – Important Cisco Commands

show ip route

show ip bgp

show ip bgp <destination-prefix>

show ip bgp neighbors

show ip route <destination-IP>

traceroute <destination-IP>

ping <destination-IP>

clear ip bgp <neighbor-IP> soft

Key Finding

Failed Protocol: BGP (Border Gateway Protocol)

Failure: Invalid BGP route advertisement / route leak

Main Evidence: Repeated AS hops in traceroute

Investigation: Examine BGP routing tables and AS-PATH

Solution: Withdraw/filter the invalid route and restore the correct BGP advertisement

Verification: Ping, traceroute, show ip route, and show ip bgp